Figure 5. Cytosolic transport slows Wave-pinning drift and activity-imbalance relaxation

Weak-input physical drift is suppressed beyond the geometric baseline in Wave-pinning, while drift predictions fail quantitatively for Goryachev. Raw heading distributions and strong-input patch statistics expose the limits of treating every change as translation of one coherent domain.

A, Physical drift diffusivity Ds relative to the smallest ring at the same f, against relative circumference. Solid curves and circles are retained native trajectory estimates; dotted curves are the deterministic susceptibility prediction with shared translation and full input covariance. All designed f ≤ 0.01 with multiple-domain fraction ≤ 0.05 enter; color intensity indexes f (10^{-4}–10^{-2}). Vertical bars are retained seed-bootstrap 95% CIs. The geometric null is 1. Goryachev has no qualified drift bridge and is explicitly unavailable here, rather than treated as a drifting domain.

B, Predicted versus observed local α, pooled over eligible f ≤ 0.01, with a single circle symbol throughout and model color. Prediction comes from deterministic native susceptibility and the same adjacent-size neighborhoods as the observed α; no noisy-data fitting or pooling average is applied. Vertical bars are observed seed-bootstrap 95% CIs; the diagonal is equality. Eligibility requires a defined native mean state, linear-response validation, resolved observed α and neighborhood multiple-domain fraction ≤ 0.05. Goryachev contributes no eligible point, a quantitative failure retained in B_alpha_bridge.csv.

C, Drift plus counted takeover angular loss divided by observed heading decorrelation, at every designed input and size. For each counted takeover whose angle is unavailable, $1 - \cos(\Delta\theta)$ is bounded independently in [0,2]. Vertical segments span the resulting partial-identification interval, not a confidence interval; circles mark its geometric midpoint when positive finite endpoints exist. Gray band is 0.85–1.15. Open triangles with arrows show one-sided intervals or portions extending past the displayed range. Light to dark denotes C20–C60 (Spring 1–3). Missing observed rates cannot supply a ratio. If drift is missing its nonnegative lower bound can be used, but the upper limit is unbounded.

D, Weak-input distributions show model-specific cores and tails: Wave-pinning's lag-scaled core broadens while its outer width changes less from 4τ to 16τ; Holmes broadens in both core and outer quantiles; Spring's core broadens while its outer quantile is nearly stable from 4τ to 16τ. Goryachev C20 has a narrow core with a growing broad tail (raw $|\Delta s| > 1.5$: 4%, 13%, 22% at τ, 4τ, 16τ), whereas C60 has no mass beyond 1.5, high coherence and no counted switches, consistent with suppression within the observed window. The plotted quantity is raw heading-change probability density at lags τ (blue solid), 4τ (brown dashed) and 16τ (black dotted), with no coherence, contrast or track-identity cut. Weak input near the C20 transition is f = 0.03, 0.02, 0.005 and 0.005 in Wave-pinning, Holmes, Spring and Goryachev; strong input is f = 0.3 for all. Facets separate model, input and C20/C60 (Spring 1/3). Signed Δθ is the principal difference in arg(P), and physical displacement is mean endpoint circumference × Δθ/(2π). Binning is in physical displacement before x and probability density are transformed by $1/\sqrt{\text{lag}}$ and $\sqrt{\text{lag}}$ respectively. The signed x axis is linear; y is logarithmic. τ = 50, Spring 5, in native time units. Zero-count bins are omitted without a pseudocount. All retained 96–384 independent

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seeds per condition enter. At strong input, Wave-pinning, Holmes and Spring spread over most of the wrapped displacement range by 4τ–16τ, so their rescaled outer widths narrow at long lag. Goryachev C60 continues to spread across lags but has weak coherence. These phase changes are not uniquely identifiable as domain relocations.

E, Strong native input produces a Wave-pinning patch field with approximately extensive patch count and residual size dependence in local statistics. Top facets show the first 2000 recorded time units after burn-in at input fraction 0.3, C20/C40/C60, seed 0; grayscale is native active concentration from 0 to 1.5. Lower facets show mean patch density (count divided by circumference), mean physical patch width, and RMS physical patch speed at input fractions 0.1 and 0.3. Patches exceed active concentration 0.65 and span at least 10 native grid cells. Speed is centroid displacement divided by the 5-unit frame interval, between mutually best overlapping regions; unmatched and ambiguous comparisons are excluded. Points are means across 64 independent seeds with 95% seed-bootstrap intervals; widths and matched speeds first pool samples within each seed. Checks at thresholds 0.55 and 0.75 retain the observed directions of size dependence and are documented in the scientific readme. No birth or death statistic is inferred. Patch count rises substantially with circumference while density, width and speed retain smaller systematic size effects; the results therefore do not support an explanation based on exactly unchanged local patch geometry. The inactive-diffusion perturbation in F tests the transport contribution despite these residual local changes.

F, Fast cytosol removes most of Wave-pinning’s strong-input size advantage: mode-1 α changes from 0.90 to 0.36 at f = 0.1 and from 0.46 to 0.02 at f = 0.3; raw-heading α changes from 0.72 to 0.21 and from 0.41 to −0.03, respectively. These headings describe orientation of whole-ring activity imbalance, rather than persistence of an established polarity domain. The plotted rate γ(q) comes from each seed's normalized autocorrelation of the complex Fourier coefficient $a_n(t)$, defined below, at the sampled lag nearest correlation 0.5; q = 2πn/C. Modes 1–16 are shown when every seed resolves a rate beyond the 5-unit frame interval. Native inactive diffusivity 10 is solid; fast diffusivity 1000 is dashed; C20/40/60 follow a light-to-dark ramp. Each point is the geometric mean across 64 seeds; shading gives 95% percentile seed-bootstrap intervals. Open upward triangles with arrows mark first-lag resolution bounds. Fast-cytosol runs alter only inactive diffusivity and are mechanism tests. The native-minus-fast mode-1 size exponent is 0.544 [0.500, 0.592] at input fraction 0.1 and 0.438 [0.425, 0.452] at 0.3; brackets are 95% paired-seed bootstrap intervals across the three sizes. The Holmes pilot check gives smaller but resolved differences, 0.081 [0.046, 0.117] and 0.081 [0.076, 0.086], respectively. Holmes retains positive mode-1 exponents after fast diffusion (0.82 to 0.74 at f = 0.1; 0.24 to 0.16 at f = 0.3): its smaller transport effects are resolved, while the remaining mode scaling is unexplained.

Physical polarity drift, counted track switching, raw activity heading, complex Fourier-mode relaxation and migration velocity memory are distinct quantities. Fast inactive diffusion is a mechanistic perturbation; all native comparisons retain native local kinetics.

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$$a_n(t) = \frac{1}{C}\int_0^C u(s,t)e^{-iq_n s}\,\mathrm{d}s, \qquad q_n = \frac{2\pi n}{C} \tag{5.1}$$

Complex spatial Fourier coefficient of active concentration u on a ring of physical circumference C; n is the integer mode index and q_n its physical wavenumber.